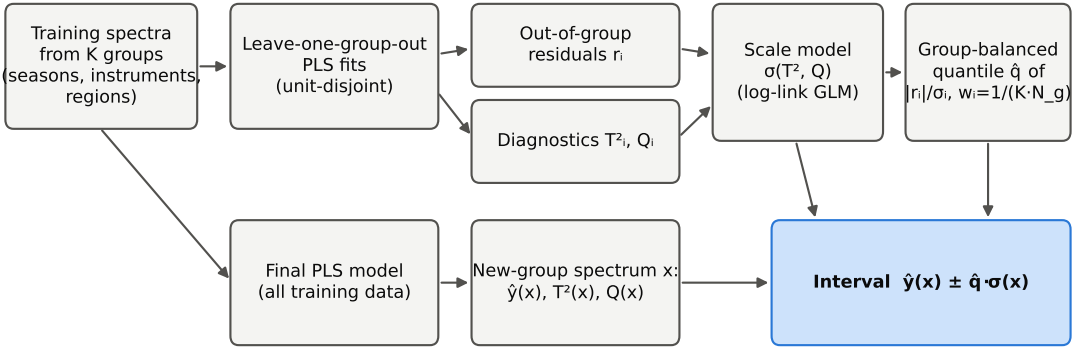
Training spectra from K groups (seasons, instruments, regions)
Leave-one-group-out PLS fits (unit-disjoint)
Out-of-group residuals r_i
Diagnostics T^2_i, Q_i
Scale model $\sigma(T^2, Q)$ (log-link GLM)
Group-balanced quantile $\hat{q}$ of $|r_i|/\sigma_i$, $w_i=1/(K\cdot N_g)$
Final PLS model (all training data)
New-group spectrum x: $\hat{y}(x)$, $T^2(x)$, $Q(x)$
Interval $\hat{y}(x) \pm \hat{q}\cdot\sigma(x)$